

UNIT IV

1. Compare and contrast Dynamic RAM with Static RAM. (APRIL 2011)
(NOV 2011)

Integrated circuit RAM units are available in two operating modes: *static* and *dynamic*.

Static RAM (SRAM) consists essentially of internal latches that store the binary information.

- The stored information remains valid as long as power is applied to the unit.
- SRAM is easier to use and has shorter read and write cycles.

Dynamic RAM (DRAM) stores the binary information in the form of electric charges on

- capacitors provided inside the chip by MOS transistors.
- The stored charge on the capacitors tends to discharge with time, and the capacitors must be periodically recharged by *refreshing* the dynamic memory.
- Refreshing is done by cycling through the words every few milliseconds to restore the decaying charge.
- DRAM offers reduced power consumption and larger storage capacity in a single memory chip.

2. Draw the block diagram of memory unit. (NOV 2011)

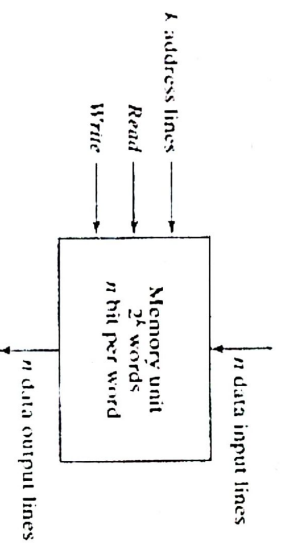


Figure: Block diagram of a memory unit

3. Write the steps taken for the purpose of transferring a new word to be stored in memory. (NOV 2012)

The steps that must be taken for the purpose of transferring a new word to be stored into memory are as follows:

1. Apply the binary address of the desired word to the address lines.
 2. Apply the data bits that must be stored in memory to the data input lines.
 3. Activate the write input.
- The memory unit will then take the bits from the input data lines and store them in the word specified by the address lines.

4. Write the steps taken for the purpose of transferring a stored word out of memory. (NOV 2012)

The steps that must be taken for the purpose of transferring a stored word out of memory are as follows:

1. Apply the binary address of the desired word to the address lines.
2. Activate the read input

5. Define CPLD. (APRIL/MAY 2012)

Complex Programmable Logic Device (CPLD), which is a collection of individual PLDs on a single integrated circuit. A programmable interconnection structure allows the PLDs to be connected to each other in the same way that can be done with individual PLDs.

6. Define FPGA. (APRIL/MAY 2012)

A field-programmable gate array (FPGA) is a VLSI circuit that can be programmed at the user's location. A typical FPGA consists of an array of millions of logic blocks, surrounded by programmable input and output blocks and connected together via programmable interconnections. A typical FPGA logic block consists of lookup tables, multiplexers, gates, and flip-flops. A lookup table is a truth table stored in an SRAM and provides the combinational circuit functions for the logic block.

7. Define the term PAL. (APRIL 2013)

The PAL is a programmable logic device with a fixed OR array and a programmable AND array. Because only the AND gates are programmable, the PAL is easier to program than, but is not as flexible as, the PLA.

8. What is Memory Unit? Write the two types of memory. (APRIL 2013 - 1T)

A memory unit is a collection of cells capable of storing a large quantity of binary information. There are two types of memories that are used in digital systems: *random access memory* (RAM) and *read only memory* (ROM).

9. Difference between RAM and ROM

- RAM is Random Access Memory, while ROM stands for Read Only Memory.
- RAM is volatile and is erased when the computer is switched off. ROM is non-volatile and generally cannot be written to.
- RAM is used for both read and write while ROM is used only for reading.
- RAM needs electricity to flow to retain information while ROM is permanent.
- RAM is analogous to a blackboard on which information can be written with a chalk and erased any number of times, while ROM is permanent and can only be read. One example is BIOS (basic input output system) that runs when computer is switched on and it prepares disk drives and to load OS from disk processor.

10. Define P.L.A.

A programmable logic array (PLA) is a kind of programmable logic device used to implement combinational logic circuits. The PLA has a set of programmable AND gate planes, which link to a set of programmable OR gate planes, which can then be conditionally complemented to produce an output. This layout allows for a large number of logic functions to be synthesized in the sum of products (and sometimes product of sums) canonical forms.

11. Define PLD

Programmable logic device (PLD) is an electronic component used to build reconfigurable digital circuits. Unlike a logic gate, which has a fixed function, a PLD has an undefined function at the time of manufacture. Before the PLD can be used in a circuit it must be programmed, that is, reconfigured.

12. Define Random Access Memory?

The architecture of memory is such that information can be selectively retrieved from any of its internal locations. The time it takes to transfer information to or from any desired random location is always the same.

13. Define Error Detection?

An Error detecting code generates multiple parity check bits that are stored with the data word in memory. Each bit is parity over a group of bits in the data word.

14. Define Read Only Memory?

A ROM is essentially a memory device in which permanent binary information is stored. The binary information must be specified by the designer and then is embedded in the unit to form the required interconnection pattern.

15. What are the secondary variables and excitation table?

The present state variables in asynchronous sequential circuits. Excitation is the next state variables in asynchronous sequential circuits.

16. Define Race Around Condition?

In JK flip flop output is fed back to the input. Therefore change in the output results change in the output. Due to this in the positive half of the clock pulse if both J and K are high then output toggles continuously. This condition is called race around condition.

17. Define Rise time?

The time required to change the voltage level from 10% to 90% is known as rise time.

18. Define skew and clock skew?

The phase shift between the rectangular clock waveforms is referred to as skew and the time delay between the two clock pulses is called clock skew.

19. Define Setup time?

The setup time is the minimum time required to maintain a constant voltage levels at the excitation inputs of the flip flops devices prior to the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip-flops.

20. Define Hold time?

The hold time is the minimum time for which the voltage levels at the excitation inputs must remain constant after the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip flop.

UNIT - IV

Reconfigurable Digital Circuits: Types of Memories – Organization of ROM and RAM – Address Decoding – Programmable Logic Devices (PLDs) – Programmable Logic Arrays (PLAs) – Programmable Array Logic (PAL) devices – Field Programmable Gate Arrays (FPGAs) - Combinational Logic implementation using PROMs, PLAs, PALs.

11 MARKS

1. Explain with a neat diagram the PAL. (11)(NOV'12)

- The PAL is a programmable logic device with a fixed OR array and a programmable AND array.
- Because only the AND gates are programmable, the PAL is easier to program than, but is not as flexible as, the PLA. Figure 7.16 shows the logic configuration of a typical PAL with four inputs and four outputs.
- Each input has a buffer-inverter gate, and each output is generated by a fixed OR gate.
- There are four sections in the unit, each composed of an AND-OR array that is *three* wide, the term used to indicate that there are three programmable AND gates in each section and one fixed OR gate.
- Each AND gate has 10 programmable input connections, shown in the diagram by 10 vertical lines intersecting each horizontal line.
- The horizontal line symbolizes the multiple-input configuration of the AND gate.
- One of the outputs is connected to a buffer-inverter gate and then fed back into two inputs of the AND gates.
- Commercial PAL devices contain more gates than the one shown in Fig. 1.
- A typical PAL integrated circuit may have eight inputs, eight outputs, and eight sections, each consisting of an eight-wide AND-OR array.
- The output terminals are sometimes driven by three-state buffers or inverters.
- In designing with a PAL, the Boolean functions must be simplified to fit into each section.
- Unlike the situation with a PLA, a product term cannot be shared among two or more OR gates.
- Therefore, each function can be simplified by itself, without regard to common product terms.

- The number of product terms in each section is fixed, and if the number of terms in the function is too large, it may be necessary to use two sections to implement one Boolean function.

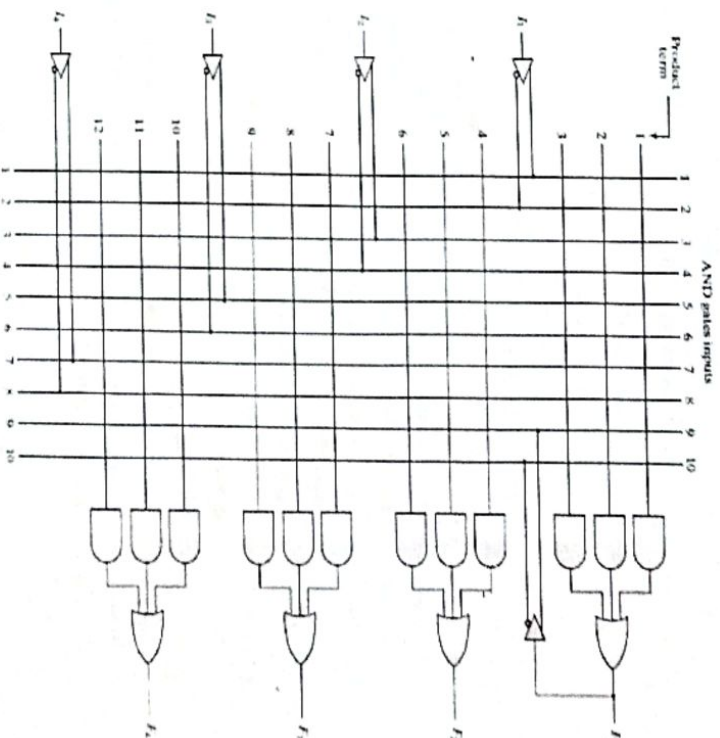


Figure 1: PAL with four inputs, four outputs, and a three-wide AND-OR structure

As an example of using a PAL in the design of a combinational circuit, consider the following Boolean functions, given in sum-of-minterms form.

Simplifying the four functions to a minimum number of terms results in the following Boolean functions:

$$w(A, B, C, D) = \sum(2, 12, 13)$$

$$x(A, B, C, D) = \sum(7, 8, 9, 10, 11, 12, 13, 14, 15)$$

$$y(A, B, C, D) = \sum(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$$

$$z(A, B, C, D) = \sum(1, 2, 8, 12, 13)$$

$$w = ABC' + A'B'CD'$$

$$x = A + BCD$$

$$y = A'B + CD + B'D'$$

$$z = ABC' + A'B'CD' + AC'D' + A'B'C'D$$

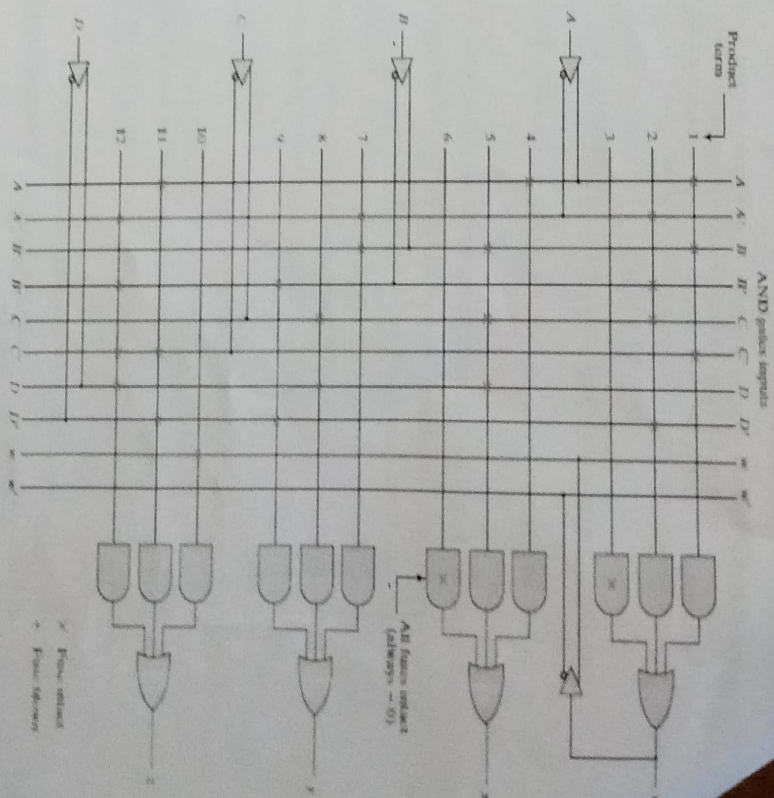
$$= w + AC'D' + A'B'C'D$$

- > Note that the function for z has four product terms.
- > The logical sum of two of these terms is equal to w .
- > By using w , it is possible to reduce the number of terms for z from four to three.
- > The PAL programming table is similar to the one used for the PLA, except that only the inputs of the AND gates need to be programmed.
- > Table 1 lists the PAL programming table for the four Boolean functions.
- > The table is divided into four sections with three product terms in each, to conform with the PAL of Fig. 1.
- > The first two sections need only two product terms to implement the Boolean function.
- > The last section, for output z , needs four product terms. Using the output from w , we can reduce the function to three terms.

Table 1: Programming Table

Product Term	AND Inputs				Outputs
	A	B	C	D	
1	1	0	1	0	$w = ABC' + A'B'CD'$
2	0	0	1	0	
3	0	1	1	0	$x = A + BCD$
4	1	1	1	0	
5	0	1	0	1	
6	1	1	0	1	$y = A'B + CD + B'D'$
7	1	0	0	1	
8	0	0	0	1	
9	0	1	0	0	
10	1	1	0	0	$z = w + AC'D' + A'B'C'D$
11	1	0	1	1	
12	0	1	1	1	

Figure 2: Fuse map for PAL as specified in Table 1



- > The fuse map for the PAL as specified in the programming table is shown in Fig. 2.
- > For each 1 or 0 in the table, we mark the corresponding intersection in the diagram with the symbol for an intact fuse.
- > For each dash, we mark the diagram with blown fuses in both the true and complement inputs. If the AND gate is not used, we leave all its input fuses intact.
- > Since the corresponding input receives both the true value and the complement of each input variable, we have $AA' = 0$ and the output of the AND gate is always 0.

- As with all PLDs, the design with PALs is facilitated by using CAD techniques.
- The blowing of internal fuses is a hardware procedure done with the help of special electronic instruments.

2. Explain with a neat diagram the PLA. (11)(NOV'11) (APR/MAY'12)

3. Discuss about PLA with diagram. (11) (APR'13)

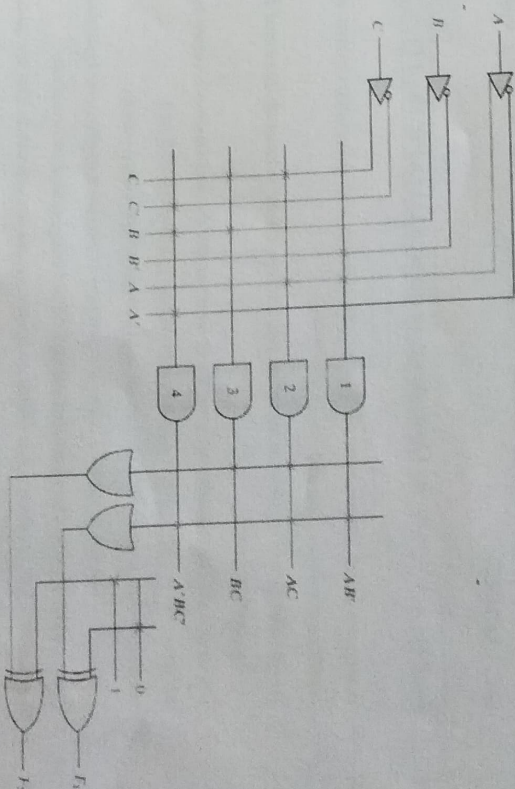
- The PLA is similar in concept to the PROM, except that the PLA does not provide full decoding of the variables and does not generate all the minterms.
- The decoder is replaced by an array of AND gates that can be programmed to generate any product term of the input variables.
- The product terms are then connected to OR gates to provide the sum of products for the required Boolean functions.
- The internal logic of a PLA with three inputs and two outputs is shown in Fig.1
- Such a circuit is too small to be useful commercially, but is presented here to demonstrate the typical logic configuration of a PLA
- The diagram uses the array logic graphic symbols for complex circuits.
- Each input goes through a buffer-inverter combination, shown in the diagram with a composite graphic symbol, that has both the true and complement outputs.
- Each input and its complement is connected to the inputs of each AND gate, as indicated by the intersections between the vertical and horizontal lines.
- The outputs of the AND gates are connected to the inputs of each OR gate.
- The output of the OR gate goes to an XOR gate, where the other input can be programmed to receive a signal equal to either logic 1 or logic 0.
- The output is inverted when the XOR input is connected to 1.
- The output does not change when the XOR input is connected to 0.
- The particular Boolean functions implemented in the PLA of Fig. 1 are

PLA Programming Table

Product Term	Inputs			Outputs	
	A	B	C	f_1 (T)	f_2 (C)
AB'	1	0	—	1	—
AC	—	—	1	1	1
BC	—	1	1	—	1
$A'BC'$	0	1	0	1	—

$$F_1 = AB' + AC + A'BC'$$

$$F_2 = (AC + BC)'$$



- In the above table, for each product term, the inputs are marked with 1, 0, or — (dash).
- If a variable in the product term appears in the form in which it is true, the corresponding input variable is marked with a 1.
- If it appears complemented, the corresponding input variable is marked with a 0
- If the variable is absent from the product term, it is marked with a dash.

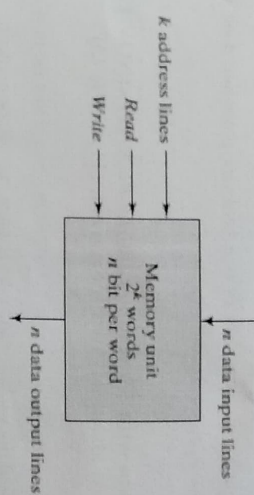
- The size of a PLA is specified by the number of inputs, the number of product terms, and the number of outputs.
- A typical integrated circuit PLA may have 16 inputs, 48 product terms, and eight outputs.
- For n inputs, k product terms, and m outputs, the internal logic of the PLA consists of n buffer-inverter gates, k AND gates, m OR gates, and m XOR gates.
- There are $2n * k$ connections between the inputs and the AND array, $k * m$ connections between the AND and OR arrays, and m connections associated with the XOR gates.

4. Explain in detail the 4×3 RAM memory. (11)(NOV'11)

5. Explain in detail about Memory organization and Memory operations. (11) (APR'12)

6. Explain RAM with its Operations in detail. (11) (APR'13 - 17)

- A memory unit is a collection of storage cells, together with associated circuits needed to transfer information into and out of a device.
- The architecture of memory is such that information can be selectively retrieved from any of its internal locations.
- The time it takes to transfer information to or from any desired random location is always the same—hence the name random access memory, abbreviated RAM.
- A memory unit stores binary information in groups of bits called *words*.
- A memory word is a group of 1's and 0's and may represent a number, an instruction, one or more alphanumeric characters, or any other binary-coded information.
- A group of 8 bits is called a *byte*. Most computer memories use words that are multiples of 8 bits in length.
- Thus, a 16-bit word contains two bytes, and a 32-bit word is made up of four bytes.
- Each word in memory is assigned an identification number, called an *address*, starting from 0 up to $2^k - 1$, where k is the number of address lines.
- The capacity of a memory unit is usually stated as the total number of bytes that the unit can store.



- Consider, for example, a memory unit with a capacity of 1K words of 16 bits each. Since $1K = 1,024 = 2^{10}$ and 16 bits constitute two bytes, we can say that the memory can accommodate $2,048 = 2K$ bytes.
- Figure shows possible contents of the first three and the last three words of this memory.
- Each word contains 16 bits that can be divided into two bytes. The words are recognized by their decimal address from 0 to 1,023.
- The equivalent binary address consists of 10 bits. The first address is specified with ten 0's; the last address is specified with ten 1's, because 1,023 in binary is equal to 1111111111.
- A word in memory is selected by its binary address. When a word is read or written, the memory operates on all 16 bits as a single unit.
- The $1K * 16$ memory of Fig. 7.3 has 10 bits in the address and 16 bits in each word.
- As another example, a $64K * 10$ memory will have 16 bits in the address (since $64K = 2^{16}$) and each word will consist of 10 bits.

Memory address			Memory content
Binary	Decimal		
000000000	0		10110101011101
000000001	1		1010101110001001
000000010	2		0000110101000110
		⋮	⋮
111111101	1021		1001110100010100
111111110	1022		0000110100011110
111111111	1023		1101111000100101

FIGURE 7.3
Contents of a 1024×16 memory

Write and Read Operations

The two operations that RAM can perform are the write and read operations.

The steps that must be taken for the purpose of transferring a new word to be stored into memory are as follows:

1. Apply the binary address of the desired word to the address lines.
2. Apply the data bits that must be stored in memory to the data input lines.
3. Activate the *write* input.

Control Input to Memory Chip

Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

The memory unit will then take the bits from the input data lines and store them in the word specified by the address lines. The steps that must be taken for the purpose of transferring a stored word out of memory are as follows:

1. Apply the binary address of the desired word to the address lines.
2. Activate the *read* input.